

Project Support Session - EasyVisa

Agenda

- Introduction to Ensemble Learning
- Bagging and Random Forests
- Boosting
- Cross Validation and Hyperparameter Tuning
- EasyVisa - Problem Statement
- EasyVisa - Full-Code/Low-Code Version
- EasyVisa - Q&A

Let's begin the discussion by answering a few questions.

Advanced Machine Learning Quiz

A company is building a machine learning model to predict customer churn. Different teams propose different approaches to improve accuracy. Which of the following approaches correctly represents an ensemble method?

A

The team builds a decision tree model and fine-tunes its hyperparameters to improve performance

B

The team trains multiple models (decision tree, linear regression, etc) and selects the one with the highest accuracy

C

The team creates multiple weak decision tree models and combines their outputs using majority voting to make a final prediction

D

The model training process is repeated multiple times on different subsets of data, and the best-performing single model is selected

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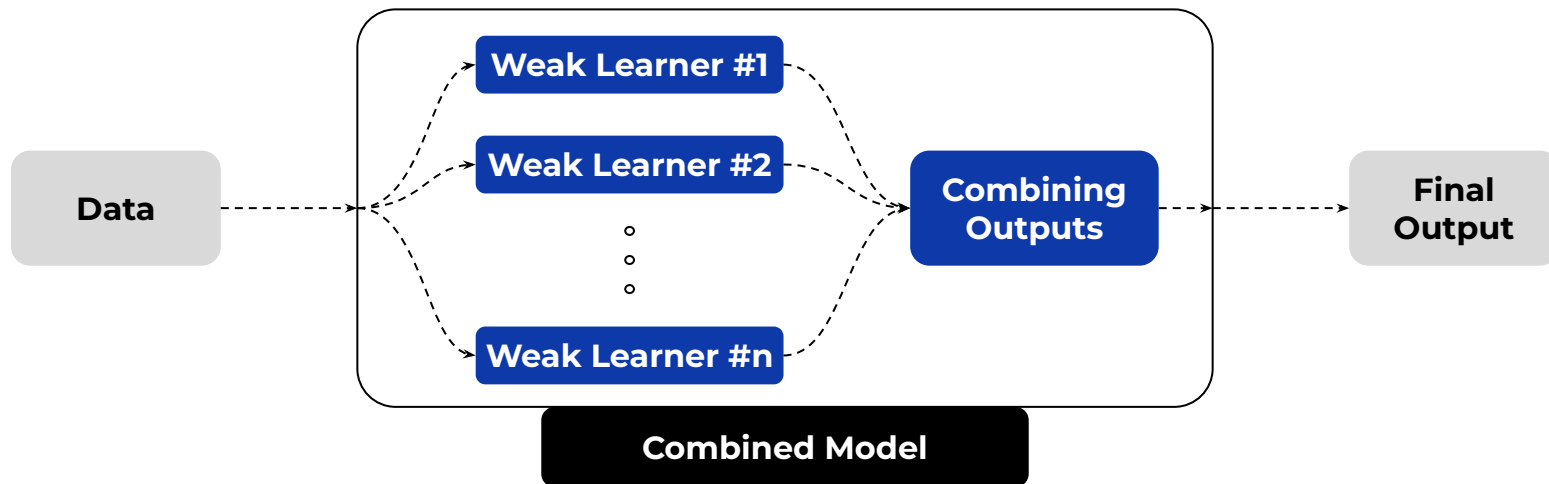
D

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Ensemble Learning

Combines the predictions of **several weak learners** built with a given learning algorithm in order to **improve generalizability** and **robustness** over a single estimator

The **motivation** behind ensemble is the belief that a committee of experts working **together** is more likely to be **accurate than individual experts**



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Ensemble Learning

Advantages of Ensemble Learning

Improved Accuracy and Stability

Combines multiple predictions to reduce errors and increase stability compared to individual models

Robustness to Overfitting

Balances bias and variance, making the ensemble less likely to overfit on training data

Capturing Non-linear Relationships

Ensembling combines diverse models or trees to capture complex, non-linear patterns

Noise Reduction

Combines multiple outputs to minimize the impact of outliers or noisy data points on predictions

Advanced Machine Learning Quiz

How does Random Forest use features/variables in a dataset when making splits during model training?

A

It considers all features at every split

B

It randomly selects a subset of features at each split

C

It randomly selects a subset of features for each decision tree

D

It uses only one feature for the entire tree

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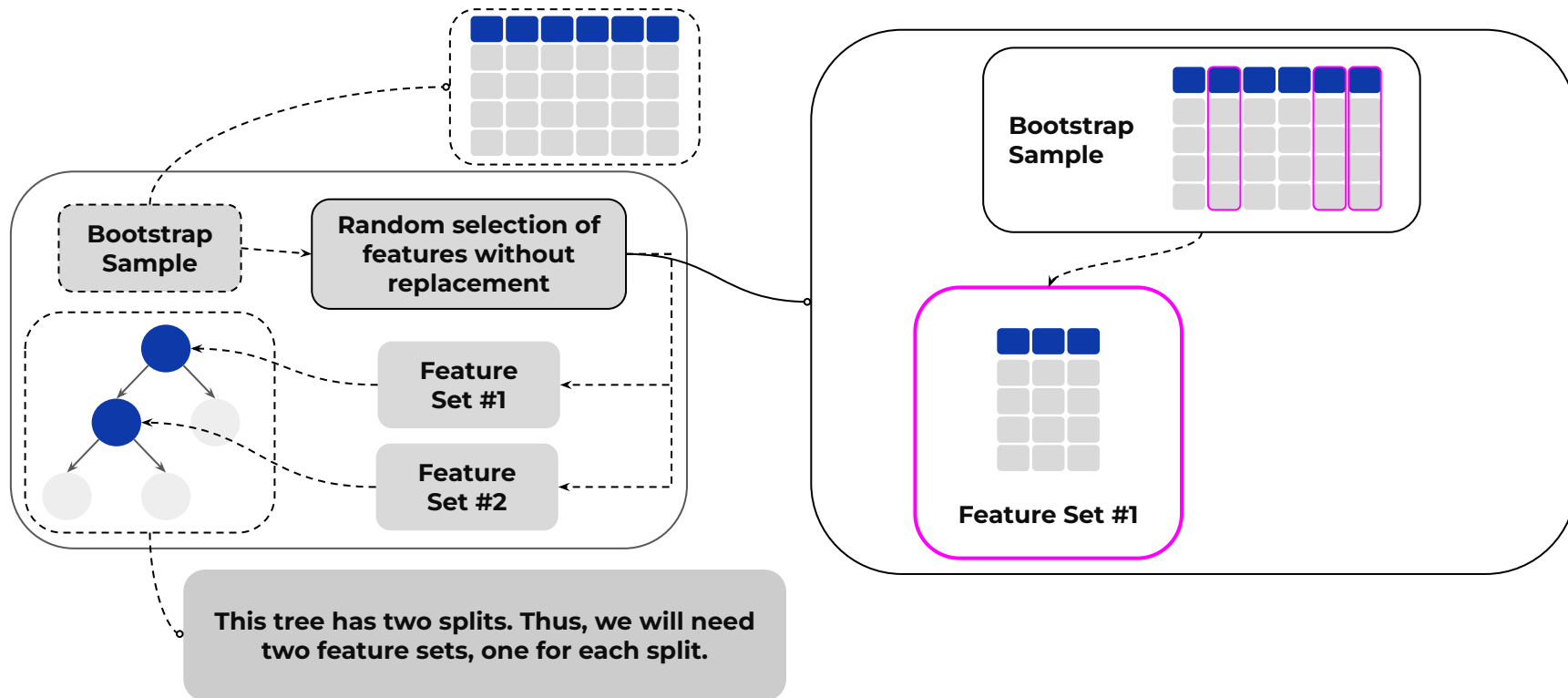
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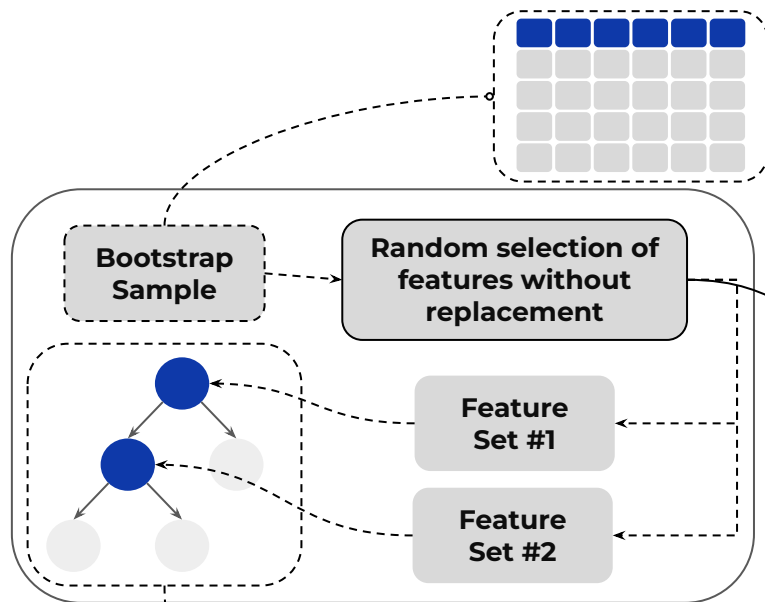
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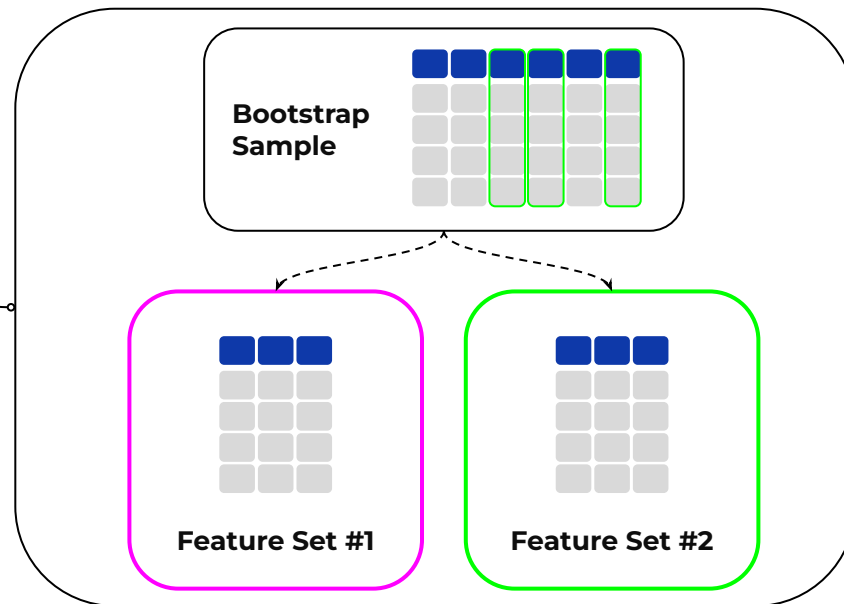
Feature Selection in Random Forest



Feature Selection in Random Forest



This tree has two splits. Thus, we will need two feature sets, one for each split.



The dependent variable will be included in both the splits

Advanced Machine Learning Quiz

What is a key differentiation of boosting algorithms from bagging algorithms?

A

Boosting trains all models in parallel

B

Boosting decreases model variance

C

Boosting increases model interpretability

D

Boosting trains all models sequentially to improve upon the wrong predictions of the previous weak learner

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Bagging V/S Boosting

Bagging

Creates different subsets of data using random sampling with replacement to independently train each weak learner

All weak learners are independent and can be trained in parallel

Aims to decrease variance, not bias

E.g, Random Forests

Boosting

Trains the weak learners sequentially on the entire dataset, reweighting misclassified data points to focus on reducing errors iteratively

All weak learners are dependent on each other and have to be trained sequentially

Aims to decrease bias, not variance

E.g, AdaBoost, Gradient Boost

Advanced Machine Learning Quiz

In AdaBoost, how are the weights of misclassified samples adjusted?

A

Increased

B

Decreased

C

Randomly assigned

D

Remains unchanged

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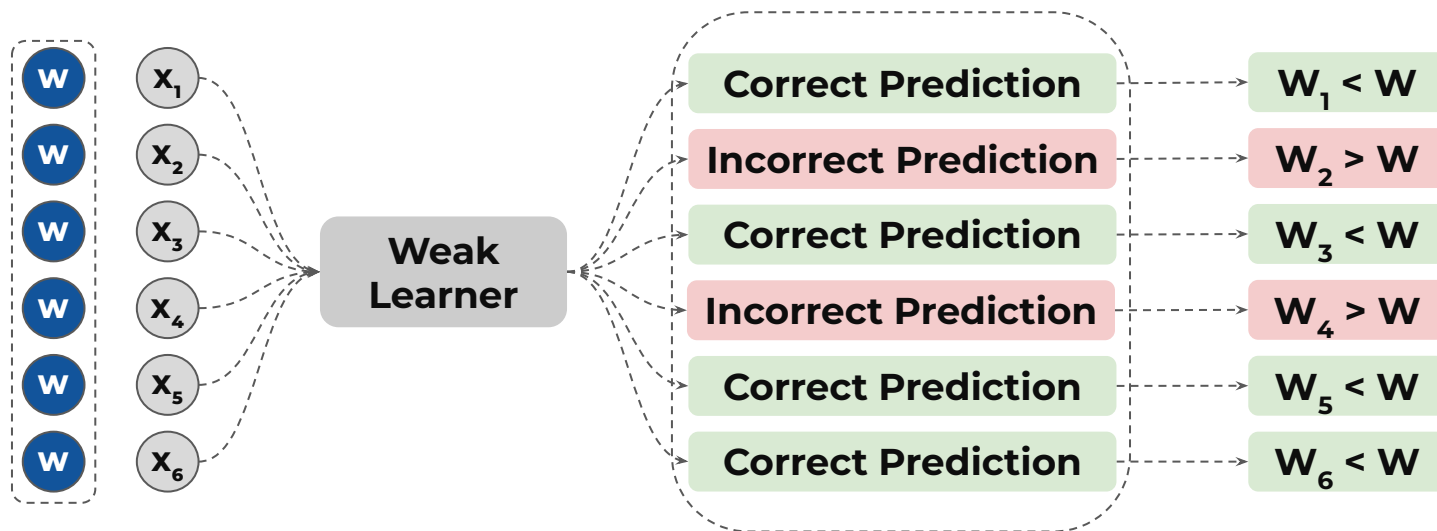
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AdaBoost - Weight Updates

Weights are **increased** for the **misclassified instances** and **decreased** for the **correctly classified instances**



Advanced Machine Learning Quiz

What role does the learning rate play in Gradient Boosting Models?

A

Adjusts the depth of trees

B

Controls the contribution of each tree to the final ensemble

C

Defines the number of weak learners to be trained

D

Determines the fraction of data used

Advanced Machine Learning Quiz

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Learning Rate in Gradient Boost

Learning rate (η) is a **small positive number** that **controls the contribution of each weak learner** to the ensemble

It controls the **magnitude of the steps** taken by the ensemble towards **an optimal solution**

0.0

η

∞

$$\hat{Y}_i = \hat{Y}_{i-1} + \eta \cdot h_i$$

η is **generally** set to smaller values (**0.01 to 0.2**) to take small steps towards the optimal solution and avoid overfitting

The **learning rate** is **multiplied** with the **predicted residual (h_i)** from each weak learner **added to the final ensemble**, thereby **dictating their contribution** by **adjusting the magnitude** of the predicted residual

Advanced Machine Learning Quiz

What is the primary difference between the ways Gradient Boost and AdaBoost improve their performance with each iteration?

A

Gradient Boosting minimizes the residual errors, while AdaBoost updates sample weights

B

Gradient Boosting updates sample weights, while AdaBoost minimizes residual errors

C

Both methods update sample weights

D

Both methods minimize residual errors

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AdaBoost V/S Gradient Boost

AdaBoost

Begins with equal importance for all data points

After training, the algorithm increases importance of misclassified instances

Subsequent learners focus more on these challenging instances

The final model combines weak learners by assigning higher importance to learners that perform well on challenging instances

Gradient Boost

Begins with an initial learner that makes simple predictions, like log(odds) value or a simple decision tree

After training, the algorithm focuses on minimizing residuals (difference between observed and predicted values)

Subsequent learners focus on learning patterns in the residuals from the previous learners

The final model combines weak learners by adding them sequentially, with each new learner reducing the errors made by the previous ones

Advanced Machine Learning Quiz

What does a larger value of k in k -fold cross-validation typically imply in terms of computational complexity?

A

Increased training time

B

Reduced memory usage during training

C

Faster cross-validation process

D

Lower computational cost per fold

Advanced Machine Learning Quiz

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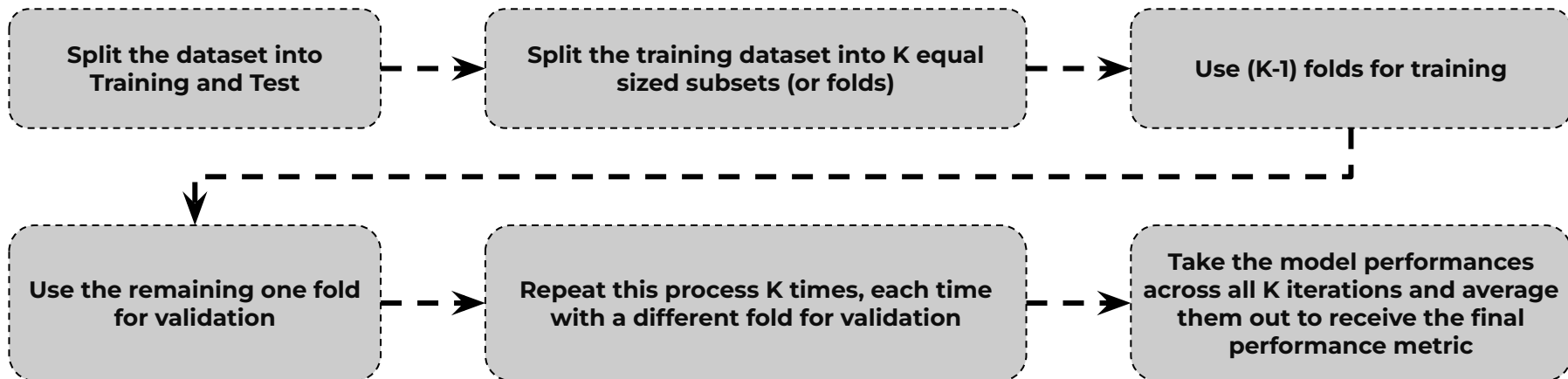
Lower computational cost per fold

K-Fold Cross Validation

A **resampling technique** used to **evaluate the performance** of an ML model

Helps assess how well a **model generalizes to unseen data**

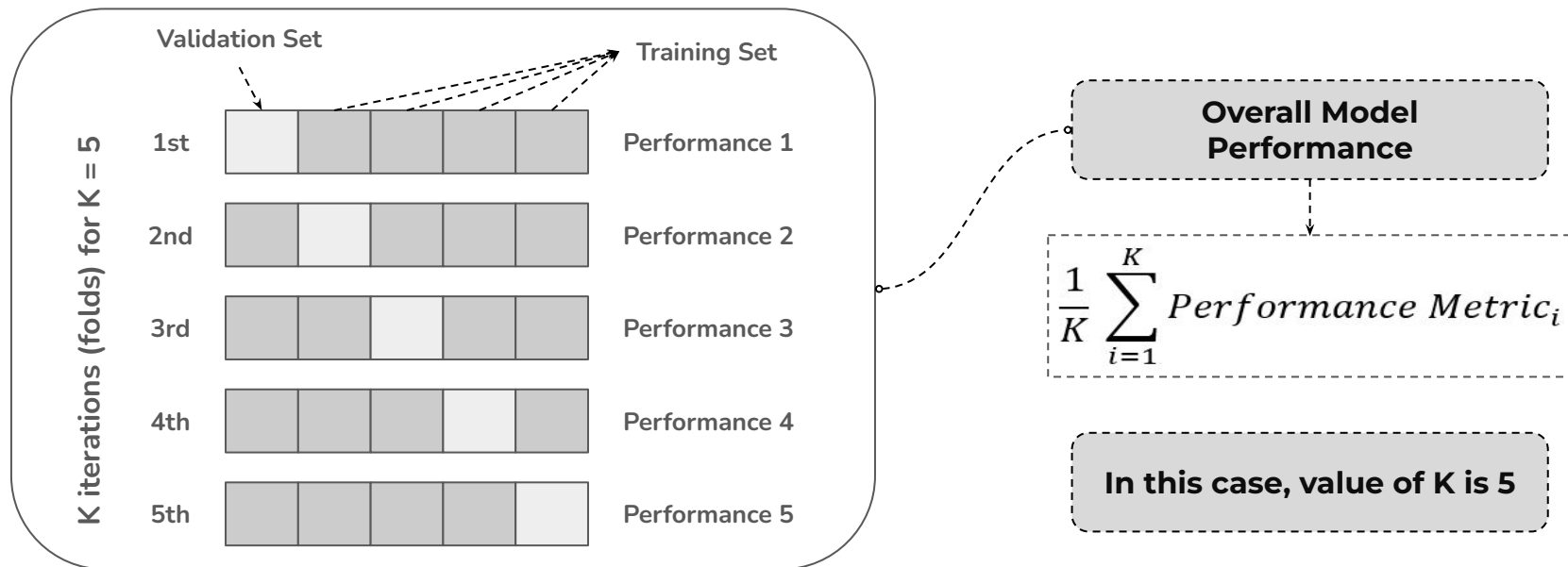
Steps for K-Fold Cross Validation



K-Fold Cross Validation

Model has to be **trained K times**, each time with a **different training set**

Higher value of K results in more training runs, and **increased training time**.



Advanced Machine Learning Quiz

You are working on a machine learning project where you need to fine-tune the hyperparameters of a model to achieve the best performance. You have a dataset with a moderate size, and you are considering using a hyperparameter optimization method. Which of the following scenarios would be most appropriate for using Grid Search?

A

The computation time is not a concern, and you need to exhaustively search over all possible combinations

B

The computation time is a major concern, and you do not need to evaluate all combinations

C

You need fast results with limited computation resources

D

Perform model selection across multiple algorithms without hyperparameter tuning

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Perform model selection across multiple algorithms without hyperparameter tuning

An algorithm used to find the **optimal set of hyperparameters** for a **model** from the **provided search space**

Consider the following set of Random Forest hyperparameters and their values to check

```
{  
    "n_estimators": [50, 75, 100],  
    "max_depth": [4, 5]  
}
```

Following combinations of hyperparameters will be tried out

n_estimators	50	50	75	75	100	100
max_depth	4	5	4	5	4	5

Advanced Machine Learning Quiz

Which of the following can cause data leakage in a machine learning model?

A

Filling missing values in the training data with attribute means computed from the training data

B

Filling missing values in the test data with attribute means computed from the test data

C

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D

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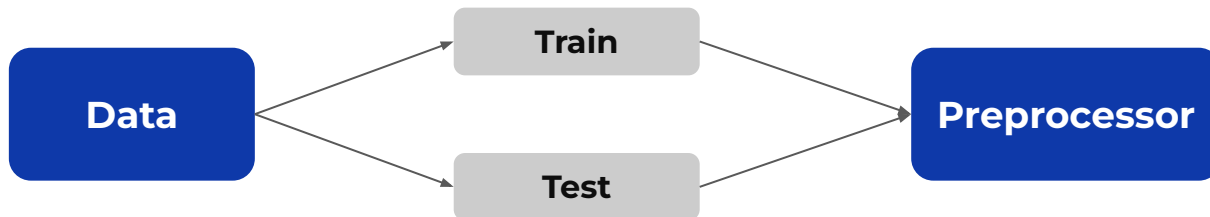
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Data Leakage



If **preprocessing** (e.g., scaling, encoding, missing value imputation) is done **before splitting**, the test set will contain **information derived from the training dataset**, potentially **leading to artificially high performance**.

As the **test data is unseen data**, we **cannot use its attributes** to preprocess the training data.

EasyVisa - Business Context

Business communities in the United States are experiencing high demand for human resources but consistently face challenges in identifying and attracting the right talent to stay competitive. Companies seek hardworking, talented, and qualified individuals, both locally and internationally.

The Immigration and Nationality Act (INA) allows foreign workers to work in the US on either a temporary or permanent basis, while protecting US workers by ensuring employers comply with statutory requirements when hiring foreign workers. These immigration programs are administered by the Office of Foreign Labor Certification (OFLC), which processes job certification applications for employers demonstrating a shortage of available US workers willing to work at wages that meet or exceed those for the intended employment area.

EasyVisa - Objective

In FY 2016, the OFLC processed 775,979 employer applications for 1,699,957 positions, marking a nine percent increase in processed applications compared to the previous year. This growing volume of applications necessitates a Machine Learning-based solution to streamline the process.

As a data scientist at EasyVisa, you are tasked with analyzing data to facilitate visa approvals and recommending suitable profiles for applicants based on the factors significantly influencing case status.

EasyVisa - Data Dictionary

The data contains the different attributes of the employee and the employer. The detailed data dictionary is given below.

Data	Data Description
case_id	ID of each visa application
continent	Information of continent the employee
education_of_employee	Information of education of the employee
has_job_experience	Does the employee has any job experience? Y= Yes; N = No
requires_job_training	Does the employee require any job training? Y = Yes; N = No
no_of_employees	Number of employees in the employer's company

EasyVisa - Data Dictionary (Contd.)

Data	Data Description
yr_of_estab	Year in which the employer's company was established
region_of_employment	Information of foreign worker's intended region of employment in the US
prevailing_wage	Average wage paid to similarly employed workers in a specific occupation in the area of intended employment. The purpose of the prevailing wage is to ensure that the foreign worker is not underpaid compared to other workers offering the same or similar service in the same area of employment.
unit_of_wage	Unit of prevailing wage. Values include Hourly, Weekly, Monthly, and Yearly.
full_time_position	Is the position of work full-time? Y = Full-Time Position; N = Part-Time Position
case_status	Flag indicating if the Visa was certified or denied

EasyVisa - Low-Code Version

For learners who aspire to be in managerial roles in the future, focusing on solution review, interpretation, recommendations, and communication with business stakeholders

Download the dataset and the *Learner Notebook - Low Code* (this is a template notebook)

Fill in the blanks in the notebook to complete and execute the code to solve the questions and perform all the tasks as per the grading rubric

Once the notebook is completely executed and necessary outputs obtained, a business presentation (using Microsoft PowerPoint, Google Slides, etc.) has to be created

EasyVisa - Low-Code Version

The presentation should contain observations, insights, and recommendations for the business problem

The presentation template provided can be referred to as a sample

Once the presentation is complete, convert the presentation to .pdf format

The presentation should be submitted as a PDF file (.pdf) and NOT as a .pptx file

Please make sure that all the sections mentioned in the grading rubric have been covered in the submission

EasyVisa - Full-Code Version

For learners who aspire to be in hands-on coding roles in the future, focusing on building solution codes from scratch

Download the dataset and the *Learner Notebook - Full Code* (this is a template notebook containing high-level steps to perform and insight-based questions)

Write necessary code to solve the questions and perform all the tasks as per the grading rubric

Clearly write down observations, insights, and recommendations for the business problem based on the analysis performed

Once the notebook is complete, download it as a **.ipynb** file and convert it to a **.html** file

EasyVisa - Full-Code Version

The notebook should be submitted as an HTML file (.html) and NOT as a notebook file (.ipynb)

The conversion can be done via one of the following ways:

Jupyter Notebook: *Download as HTML* from the **File** menu

Google Colab: Use free online tools

Please make sure that all the sections mentioned in the grading rubric have been covered in the submission

EasyVisa - Q&A



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Power Ahead!

